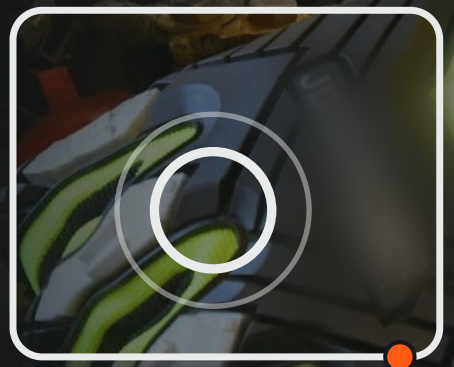
praxis

STEP 4/9

# Instructions that can finally see how you're doing

An AI coach for new maintenance technicians. It watches the work through a first-person camera and says the next step.



Seat the clamp past the hose bead

STARTUPS @ HARVARD · 20 OCT 2026 · MONISH SARAVANA

CONCEPT MOCK · STOCK PHOTO

THE PROBLEM

# A new maintenance technician needs about a year of a senior tech's time to work alone

**9–14**

months before a new technician pays for themselves, in a survey of 35 repair stations.

**$10–15k**

to hire and ramp up one maintenance technician, counting onboarding and lost productivity.

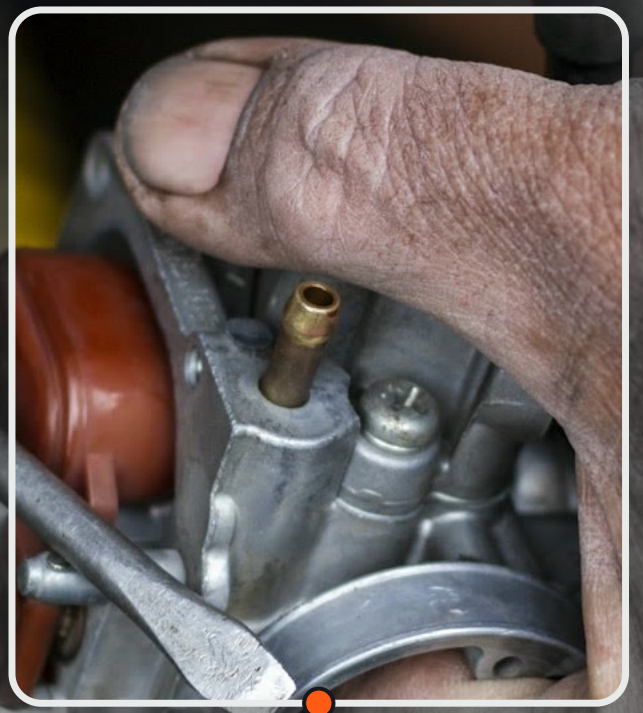
**~255,000 openings a year** for US maintenance and repair techs, and every new hire pulls a senior tech off the floor.

THE PRODUCT

# Praxis watches the work and coaches one step at a time

| | |
|---|---|
| **LOAD** | the manuals and checklists a shop already has |
| **COACH** | on a phone or glasses, one short line at a time |
| **VERIFY** | critical steps and keep the record |



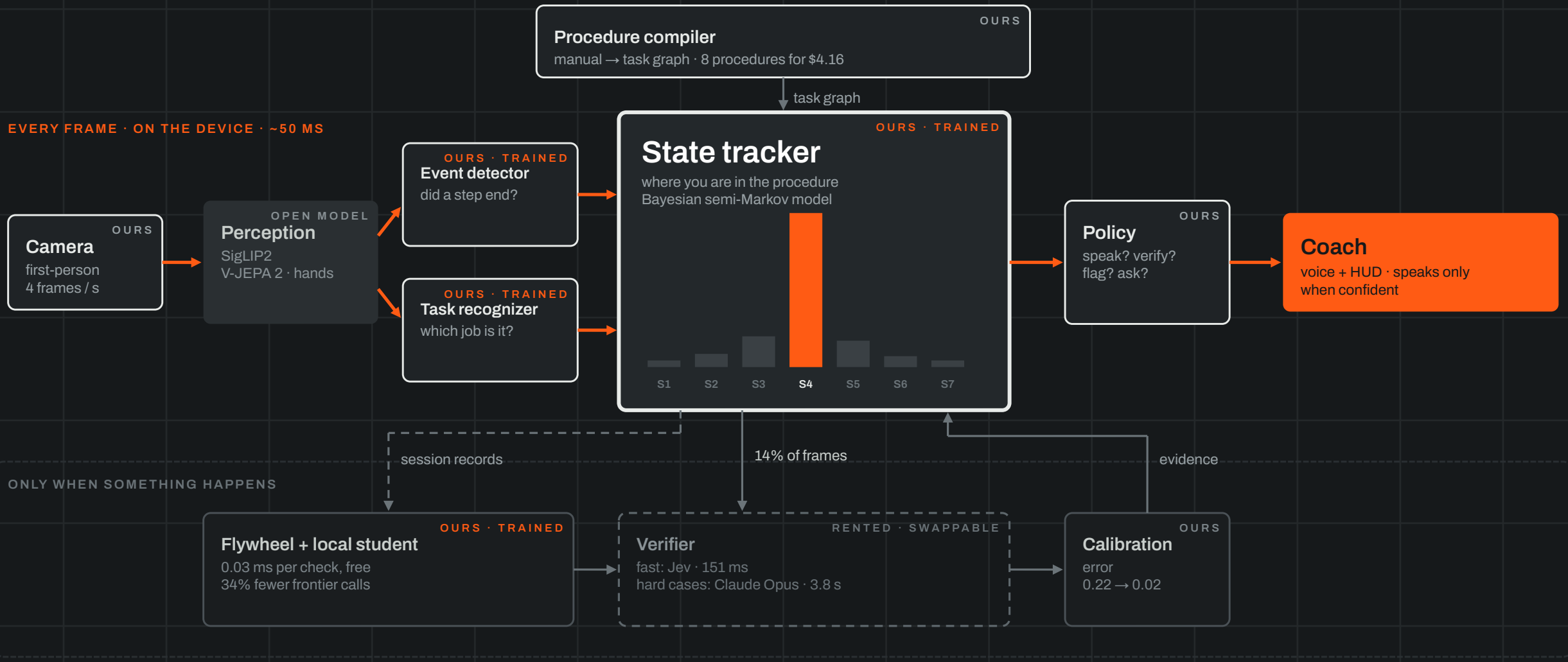


Loosen the four float-bowl screws a half turn each

CONCEPT MOCK · STOCK PHOTO

HOW IT WORKS · ENGINE v0

# The frontier model is one swappable part. The tracking engine around it is ours.



# A shop uploads the manuals it already has. Praxis turns each procedure into a coach

3 DOCUMENTS

| DOCUMENT | PAGES | SECTIONS | LIKELY PROCEDURES |
|---|---|---|---|
| **FIST 1-1 · Hazardous Energy Control Program**<br>FIST_1-1.pdf · v1 | 97 | 131 | 4 |
| **FIST 4-1A · Maintenance Scheduling for Mechanical Equipment**<br>FIST_4-1A_2020.pdf · v1 | 222 | 234 | 3 |
| **FIST 3-8 · Protective Relays: Operation, Maintenance and Field Tests**<br>FIST_3-8_2021.pdf · v1 | 125 | 97 | 0 |

1. **Upload the manuals**
   A 97-page safety manual is indexed in 0.34 s on a laptop, for $0. Its procedures are found automatically.

2. **Review the coached steps**
   Each procedure compiles for about $0.19, with cards quoted from the manual. A manager approves it first.

3. **Coach on a phone or glasses**
   Live tracking, voice control, Meta and Viture glasses connections, screen or voice-only modes.

## High Voltage Board Repair (Electric Screwdriver)

v1 · approved · live · Approved · live

Source **High-voltage board repair (electric screwdriver)** v1 · High Voltage Board Repair Procedure (Electric Screwdriver) · Steps from split call · Sent 6,569 chars of a 1-page document
Local tracking: document only · mark steps on a recording · Cost $0.26 · claude-opus-5-5 · Oct 18

WHOLE PROCEDURE

- **high voltage** HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECT... ×
- **high voltage** Observe the high voltage board for a moment ×
- **high voltage** Take the high voltage board and places it on the work area ×
- **electrical: 5 Volts** Adjust the power supply's voltage knob to set it to 5 Volts ×
- **disconnect the power** Disconnect the power supply cables ×
- **temperature: 200 degrees** Set the soldering iron temperature to 200 degrees using th... ×
- **temperature: 160 degrees** Reduce the soldering iron temperature to the minimum (1... ×
- **temperature: 160 degrees** Lower the soldering iron temperature to the minimum (160... ×
- **electrical: 5 Volts** Set the voltage knob of the power supply to 5 Volts ×

STEPS 20 · 0 safety-critical — Expand all

1. **Start recording and toggle bench lamp** — Sit at the workbench and say "Record" clearly, then wait for the b... — Safety wording in source — 1 card
2. **Press green buttons on Panel A** — Walk over to Panel A and look at it for a moment. Then press both green buttons.
3. **Place board and connect screwdriver battery** — Go back and sit at the bench, look at the high voltage board, th... — Safety wording in source — 2 cards
4. **Secure board to workbench with screwdriver**

**What the compiler saw** — 6,569 chars

Page 1 ↗

HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECTRIC SCREWDRIVER) Sit at the workbench Pronounc the voice command "Record" to start the recording; you will hear an audible signal for confirmation Turn on and off the lamp on the workbench while looking at it Leave the laboratory Enter the laboratory and closes the door Head towards Panel A and observes it for a moment Press the two green buttons on Panel A Move back to the workbench and sits down Observe the high voltage board for a moment Take the high voltage board and places it on the work area Connect the battery to the electric screwdriver Remove the screws from the workbench using the electric screwdriver Secure the board to the workbench using the electric screwdriver Observe the power supply Turn on the power supply using the socket switch Adjust the power supply's current knob until the green LED lights up Adjust the power supply's voltage knob to set it to 5 Volts Connect the power supply cables to the board's power points Watch the board for a few seconds to check if the red and green LEDs turn on Turn off the board's power supply using the socket switch and wait for the board's LEDs to turn off Set the current and voltage knobs of the power supply back to 0 Disconnect the power supply cables Remove the board from the workbench using the electric screwdriver Observe the soldering iron Turn on the soldering iron using the socket switch Set the soldering iron temperature to 200 degrees using the yellow "UP" button Grab the capacitor on the board with pliers Take the soldering iron's probe Touch the first pin of the capacitor with the soldering iron's probe for 5 seconds Touch the second pin of the capacitor with the soldering iron's probe for 5 seconds Place the pliers on the workbench Place the soldering iron's probe Position the board vertically Grab

HOW FAR ALONG · MEASURED

# It already beats a frontier model on its own. The bottleneck is maintenance data

## 13% more

accurate at tracking the step than the same frontier model called every 5 seconds

## 1/5 the cost

$0.018 vs $0.085 per minute of video, because it only asks a model when something happens

## 11 of 15

technician questions on an industrial repair job answered correctly from the company's own manuals

## 0

times confidently wrong when the view was blurred or blocked. It asks for a better view instead

THE BOTTLENECK: DATA

First-person video of real maintenance work isn't sold off the shelf. So we collect it:

- Pay trainee techs at trade schools to wear a camera on real jobs
- Label it with our own step-marking tool
- Every pilot adds more, with the customer's consent

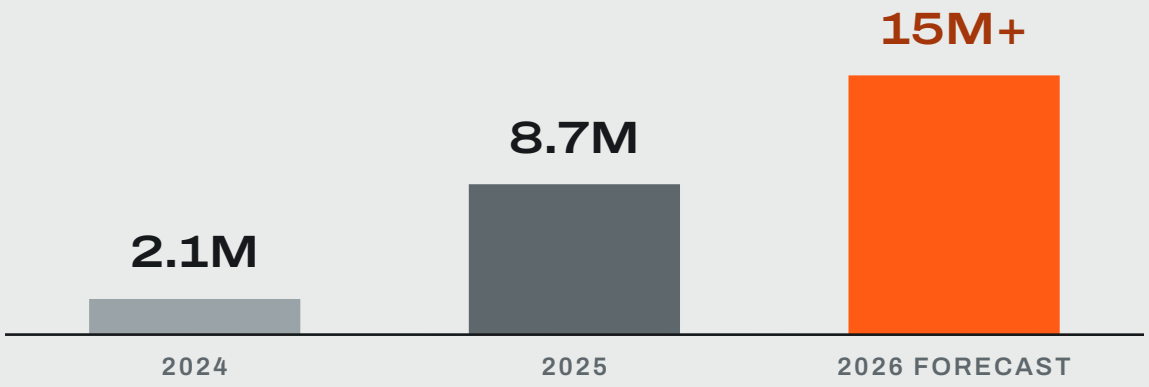
**About 80 hours of footage for $2,000, then GPU time to retrain.**

WHY NOW

# Smart glasses took off, and AI keeps getting cheaper

## AI GLASSES SHIPPED A YEAR

**7×** more glasses in two years. Every pair is a camera we can run on




## PRICE OF THE SAME AI, PER MILLION TOKENS

**1,000×** cheaper in three years, about 10× every year

2021 · GPT-3 at launch — $60

**2024 · same capability** — $0.06

So a coach that watches a whole shift is finally affordable.

**MAY 2026**
Autodesk buys MaintainX, a maintenance app, for $3.6B

**SEPT 2026**
Meta opens its glasses' camera to outside apps

**NOV 2026**
Android XR glasses reported to ship

**Q1 2027**
Viture + NVIDIA coaching glasses ship

COMPETITORS

# Rolling out a new procedure: what it takes today, and what it takes with Praxis

| | | | | |
|---|---|---|---|---|
| **Procedure apps**<br>MaintainX, Augmentir | Type the steps into the app | The tech reads them on a phone | Ticks a box when done | Nobody sees whether the step was done right |
| **Fixed cameras**<br>Retrocausal, Invisible AI | Mount a camera at one workstation | Record and label runs to train it | It watches that one station | Can't follow a tech around a machine |
| **First-person AI glasses**<br>Strivr, Viture + NVIDIA | Their team films your procedures | They train a model for your site | Coach through their glasses | Filming and training for every customer |
| **Praxis** | Upload the manual you already have | Approve the coached steps, about $0.19 each | Coach on a phone or glasses | **Verified and recorded, with no filming** |

WHY NOBODY OWNS THIS NICHE

# Meta, Google and Apple won't specialize, so we will. Their glasses become our hardware

## 15M+

AI glasses forecast to ship this year. Every pair is a camera we can run on, and we pay $0 to build it.

## Sept 2026

Meta opened its glasses' camera to outside apps. The platforms want specialists building on top.

## $0

What Meta AI and Gemini Live charge consumers to look through the camera and help. We don't sell against free.

THEY BUILD FOR EVERYONE

Glasses, cameras and one free general assistant that answers questions about a single frame.

WE BUILD FOR ONE TRADE

A coach for maintenance teams: their own procedures, step-by-step state, verified records, a pilot on site.

**That's also why we don't start with consumers.** Consumer help is their turf and they give it away free. In a maintenance shop someone pays, and the same checklist runs every month.

MARKET + PRICING · US, PER YEAR

# A $306M wedge in a $1.8B first market, in a $9B one

**TAM** — $9.2B

15.4M US workers who follow written procedures

**SAM** — $1.8B

2.47M US maintenance and repair techs

**WEDGE** — $306M

~255k new maintenance techs a year, coached

**YEAR 3** — $4.5M ARR

~3,750 trainees, about 150 teams

## HOW WE CHARGE

**A short paid pilot**
with a metric agreed up front

**$200 per trainee a month**
while a new tech is being coached, about six months

**One-time setup per procedure**
we compile and check the shop's own manuals

**$50 per tech a month, next**
verified records for the whole team

SAVINGS + GO-TO-MARKET

# A new tech costs $10–15k to hire and ramp up. We charge $1,200. One saved hour a week pays for it

## $10–15k

to hire and ramp up one maintenance tech today

## $1,200

per trainee: $200 a month for six months, after a short paid pilot

## 1 hour

a week of a senior tech's time ($44) is what a $200-a-month seat costs, so saving one hour pays for it

HOW WE GET CUSTOMERS

CUSTOMERS 1–10

### Founder-led sales

I sign the first maintenance teams and repair shops myself, as paid pilots. They pay us

→

CUSTOMERS 10–100

### Results sell it

Pilot results become case studies, and Praxis plugs into the maintenance apps shops already use

→

CUSTOMERS 100–1,000

### Partners resell it

Equipment distributors and glasses makers sell Praxis with their products

THE LONG GAME

# Every coaching session builds data nobody else has. That data is how we climb from technicians to robots

NOW · 2027

## Coach new techs

**$0.3M**

10 teams from paid pilots

NEXT · 2028

## Verified records

**$1.5M**

50 teams + records · 5×

THEN · 2029

## Our own models

**$4.5M ARR**

150 teams · better margins · 3×

LATER · 2030+

## Robots: two paths

**$13M → $30M ARR, 2030–31**

**A** Train our own robot models on the data we've collected

**B** Sell the data and our step checker to robot labs

AI LABS ALREADY PAY FOR DATA, AND IT'S GROWING

**$2B+ a year**

Mercor's run-rate selling expert data to AI labs, doubled in four months

**>$100M a year**

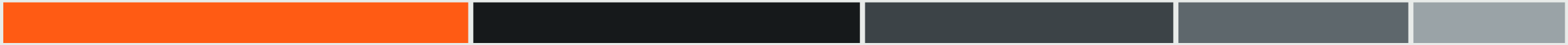
what robotics companies already spend buying real-world data

**$8.6B by 2030**

the AI training-data market, growing about 22% a year

THE ASK

# $10k buys the data and GPU time to make the models better, and puts Praxis on glasses at a first pilot



## $3,000

### GPU time

Train our models (step tracker, event detector, local checker) on more data

## $2,500

### Glasses + capture

Run Praxis on camera glasses at a pilot site

## $2,000

### Training data

Buy labeled first-person recordings of real maintenance work

## $1,500

### Inference

About 1,400 hours of live pilot coaching, at ~$1.08 an hour

## $1,000

### Company formation

So a customer can sign a paid pilot

---

TRUE BY APRIL

**Models retrained on real maintenance recordings**

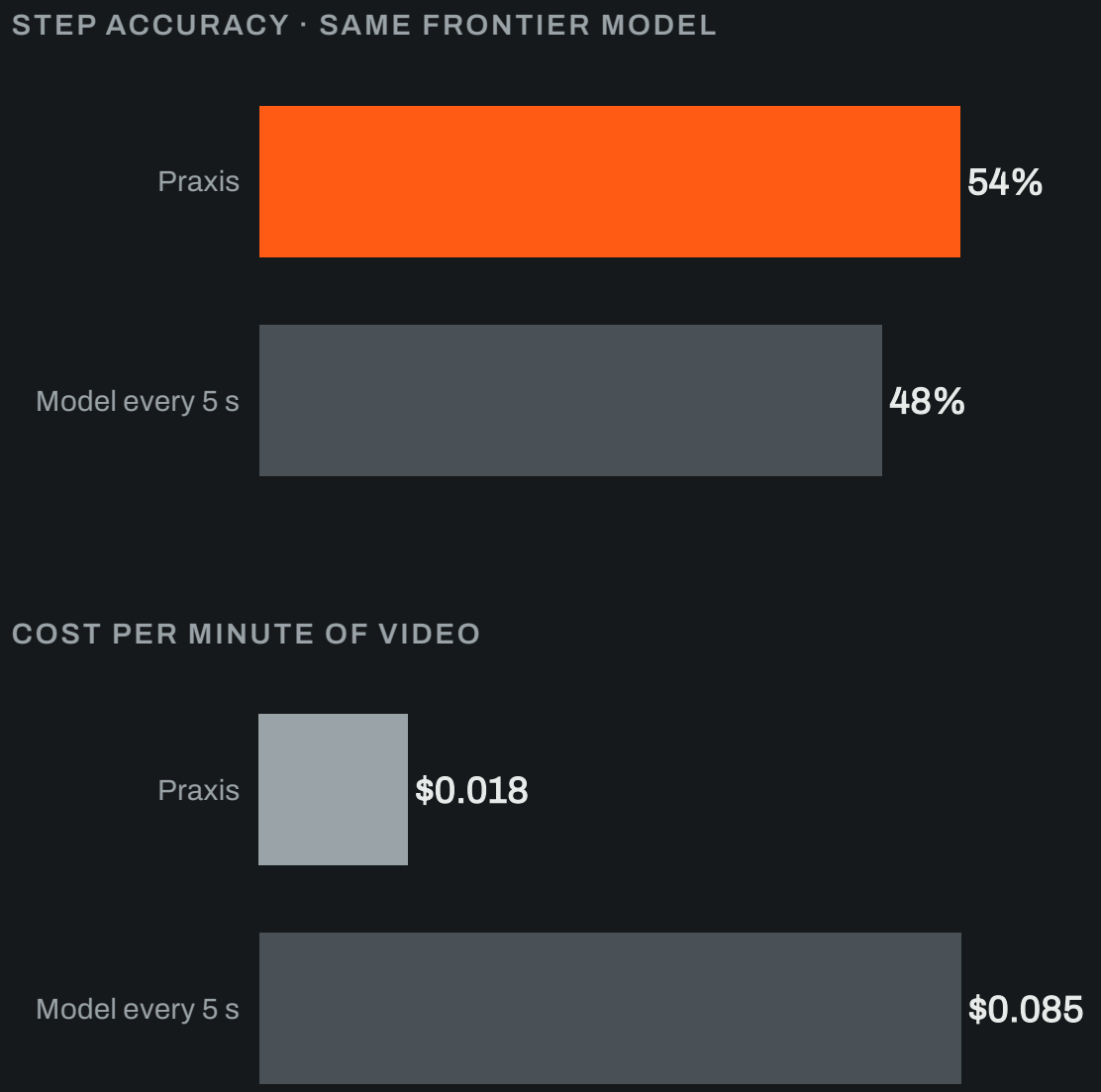
**Running on glasses at a real site**

**A first paid pilot, with a metric agreed up front**

INFO SLIDES

# Appendix


| | |
|---|---|
| A1 | Evidence: accuracy and cost |
| A2 | What we train, and why big labs won't |
| A3 | The status quo, tool by tool |
| A4 | Market math and assumptions |
| A5 | Unit economics and pricing risk |
| A6 | Learning curves |
| A7 | Training data, stated plainly |
| A8 | Competitor detail |
| A9 | Why not consumers first |
| A10 | Risks and what we do |
| A11 | Glasses and voice |
| A12 | Why me |




APPENDIX · A1

# Same frontier model, with our tracking: 13% more accurate at a fifth of the cost

**1/5**

the cost per minute of video, and more accurate.

APPENDIX · A2

# What we train in-house, and why a big AI lab won't build this

| PART | WHO | WHAT IT DOES |
|---|---|---|
| **Event detector** | OURS · TRAINED | A small temporal network that spots the moment a step ends |
| **Task recognizer** | OURS · TRAINED | Works out which procedure the tech is doing |
| **State tracker** | OURS · TRAINED | A Bayesian model of where the tech is in the procedure, with step models per procedure |
| **Local checker** | OURS · TRAINED | Learned from the frontier model; answers routine "is this step done?" checks for free |
| **Perception** | OPEN MODELS | SigLIP2, V-JEPA 2 and hand tracking turn each frame into features. Not trained by us |
| **Verifier** | RENTED | A frontier model answers hard questions on 14% of frames. Swappable in one line |

### WHY A BIG LAB WON'T

**They sell one general model.** Ours wraps any model, so their best one becomes our swappable verifier.

**The hard part is data they don't have:** first-person video of skilled work, lined up to the procedure, with verified steps.

**And a business they don't run:** each shop's manuals and records, and a pilot on site.

# The status quo: what each tool does today, and what it misses

SHADOWING

**A senior tech, side by side**

The real teacher today

MISSES About $44 an hour of senior time, one trainee at a time

MANUALS + VIDEOS

**PDFs, binders, YouTube**

Explain what should happen

MISSES Can't see what you're actually doing

PROCEDURE APPS

**MaintainX, Tulip, Augmentir**

Put the steps on a phone; collect sign-offs and photos

MISSES A checkbox isn't proof the step was done right

FIXED CAMERAS

**Retrocausal, Invisible AI**

Watch one workstation on a line

MISSES Maintenance moves around the machine

PRAXIS

Watches the work, follows the person around the machine, and runs on hardware a shop already owns.

# Market math, and the assumptions behind it

| | WHO (BLS) | MATH | PER YEAR |
|---|---|---|---|
| **TAM** | Installation, maintenance + repair 6,086,190 · electricians 757,220 · production 8,565,760 = 15.4M | 15.4M × $50 a month × 12 | ≈ $9.2B |
| **SAM** | General maintenance + repair, machinery mechanics + millwrights, HVAC = 2.47M jobs, ~255k openings a year | 255k × $200 × 6 months = $306M; 2.47M × $50 × 12 = $1.48B | ≈ $1.8B |
| **SOM (year 3)** | 1.5% of new maintenance techs ≈ 3,750 trainees a year, ~150 mid-size teams | 3,750 × $200 × 6 months | ≈ $4.5M ARR |

**Prices are hypotheses.** $200 a month for coaching anchors on Tulip's $100–250 per station; $50 a month for every-worker records is a guess

**Openings** include people changing jobs, so new-tech counts are an upper bound

**Not counted:** other countries, field service outside these groups, aviation maintenance, and the robotics upside

APPENDIX · A5

# Unit economics, and the pricing risk

| DESIGN | $ / VIDEO-MIN | $ / TRAINEE-MONTH, 2 H/DAY | MARGIN AT $200 |
|---|---|---|---|
| Frontier model every 5 s | $0.085 | about $214 | negative |
| Praxis engine today | $0.018 | about $45 | about 77% |
| Praxis + our local model | $0.008 | about $20 | about 90% |

Labeled assumptions, not sourced: if a senior tech spends 50% of their time on a new hire for 3 months, that's about $11,400 of their time; at 25% for 12 months, about $22,800. We're asking maintenance managers for their real split.

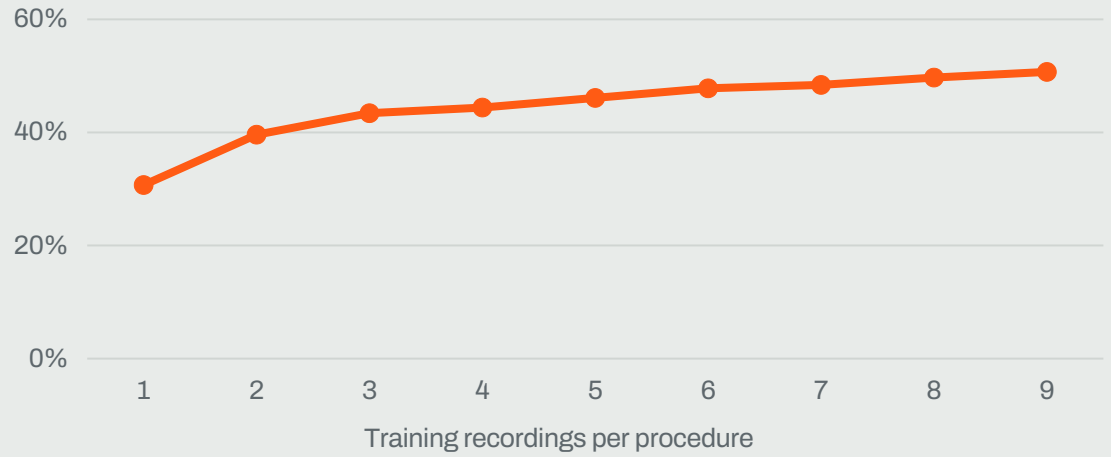
**THE RISK**

All-shift use (8 h/day) makes a flat $200 seat $1.19 an hour against our $1.08 cost: about a 9% margin. With the local model, about 60%. So price by coached hours or tiers, and move routine checks to the local model first.

# Both learning curves: step models and the local student

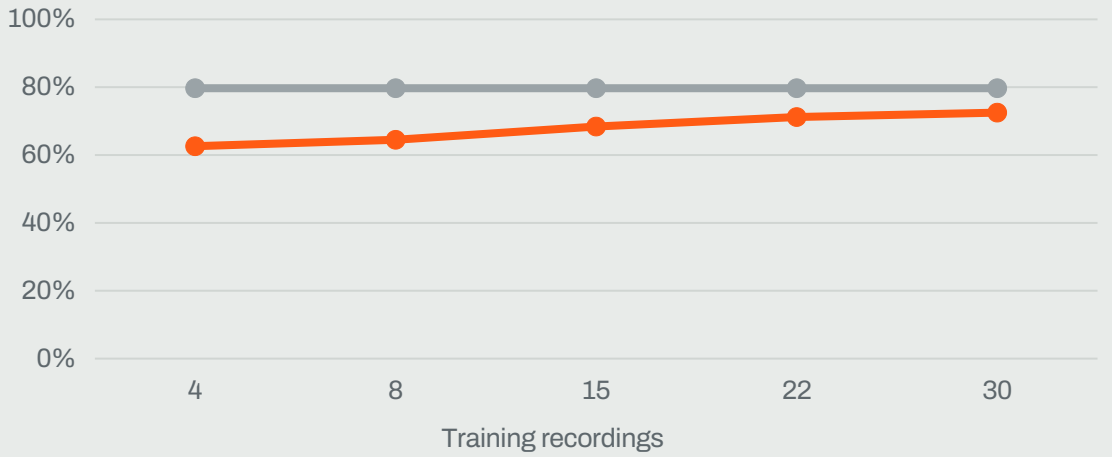
## 30.7% → 50.7%

recognizing the current step, on videos it never trained on




## 62.6% → 72.5%

checking a step is done · grey: the frontier model, 79.7%




TESTED ONLY ON RECORDINGS IT NEVER SAW · BOTH STILL RISING AT THE LARGEST SIZE WE HAVE

# Training data, stated plainly

**Train and test never overlap.** Everything learned uses only the dataset's official train-split recordings

**Small on purpose.** A weekend-scale proof: the step-change detector trained on 24 recordings, the step models on 3 per procedure, the student on checks from 30

**Tested on 48 held-out clips:** 235 min, 302 annotated steps, 170 annotated mistakes, plus procedure types never seen in training

**Design choices** were made on train-split recordings only, never on the test clips

**Labels are the dataset's,** not ours. One run per system, no confidence intervals yet

## LEARNING CURVES

**Step models:** 30.7% with 1 recording → 43.4% with 3 → 50.7% with 9 (the 3 procedures with 9)

**Student:** 62.6% with 4 recordings → 72.5% with 30; the frontier teacher is 79.7%

**Detector:** noisy, no clear trend yet. Not claimed

# Who else is here, and how we differ

| CATEGORY | EXAMPLES | US VS THEM |
|---|---|---|
| Instruction platforms | Tulip, Augmentir, Dozuki | They show the steps; we watch the work. Natural integration partners |
| Line cameras | Retrocausal, Invisible AI, Tulip Vision | Fixed at a station on a production line. Can't follow a person around a machine |
| First-person coaching | Viture Helix + NVIDIA, Strivr | Closest. Hardware-led or high-touch; no published accuracy |
| Toolkits | NVIDIA XR AI, Meta's camera toolkit, Android XR | Complements: the plumbing we build on |
| General assistants | Meta AI, Gemini Live | Answer questions about a frame. No plan, no state, no calibration |

If a big platform ships a coach: they're building our hardware and plumbing. We go deep in one vertical, with its procedures, records and data.

# Why not consumers first? Because the budget and the repeat data are in business

**$3.6B**

what Autodesk agreed to pay for MaintainX, an app that puts maintenance procedures on a phone.

**$99 → $25**

what Frontdoor's expert video help for homeowners fell to, per year, from 2023 to 2024.

**The same checklist runs on every machine, every month.** Home tasks don't repeat, and the platforms give consumer help away free. Consumers come last.

APPENDIX · A10

# The risks we see, and what we do about each

| RISK | WHAT WE DO |
|---|---|
| **Meta or Google ships a coach** | They're building our hardware and plumbing. We go deep in maintenance with its procedures, records and data, and run on their glasses |
| **Glasses aren't ready** | It works on a tablet or phone today. Glasses are one more input adapter |
| **Mistake detection is weak (4 of 17)** | Locked procedures, maintenance data and the tech saying "done" all add evidence. Measured in every pilot |
| **Privacy, unions, monitoring laws** | Fast layers run locally; only event keyframes leave (14% of frames). Coaching for the worker, aggregates for the employer. Blur next |
| **A wrong prompt on a safety-critical step** | Lockout and similar steps stay the tech's job. Praxis prompts and records them, and never approves them |
| **Depending on model vendors** | The verifier is swappable in one config line, and our local student already handles routine checks |
| **Solo founder** | The grant tests demand before hiring. First hires come after the pilot |

# Glasses and voice: what's built, and what's next

## BUILT

- Runs live in real time, with the local layers at about 50 ms per frame; model calls never block the loop
- Glasses connections for Meta and Viture + NVIDIA, plug and play
- Voice control, plus screen or voice-only modes depending on the glasses
- Praxis Studio: the engine in a browser, on a camera, an upload or a test clip

## NEXT, WITH DEVICES IN HAND

- Test end to end on Meta and Viture glasses at a pilot site
- Measure glass-to-coach latency, the biggest unknown
- A hearing-protection headset for loud plants
- Face and badge blur; a local-model tier for sensitive sites

# I've built real-time vision on messy video before, and I built this in a week

## LipTrac

Real-time lip reading from ordinary video. 60,000+ frames, 12 testers, provisional patent filed.

## 7 years

of competitive robotics, coaching teammates without seeing what they see.

## 1 week

to build the engine, a 48-video eval and a live app, solo. First commit Oct 17.

Monish Saravana · Harvard College · solo founder. Also: path planning for oil-cleanup boats from aerial imagery.